Physics of the Earth and Planetary Interiors 274 (2018) 222–231

Contents lists available at ScienceDirect

Physics of the Earth and Planetary Interiors

journal homepage: www.elsevier.com/locate/pepi

Short communication

Paleomagnetism and geochronological studies on a 450 km long 2216 Ma dyke from the Dharwar craton, southern India

E. Nagaraju[a,b,*], V. Parashuramulu[b], Anil Kumar[b], D. Srinivas Sarma[b]

[a] TSMS Jr College, Baswapur, Sangareddy Ditrict, Telangana 502293, India
[b] CSIR-National Geophysical Research Institute, Hyderabad 500007, India

ARTICLE INFO

Keywords:
Paleomagnetism
Geochronology
Dharwar craton

ABSTRACT

Paleomagnetic and geochronological studies were carried out on a ~ 450 km long (from 17 sites) N-S striking Paleoproterozoic dyke swarm exposed along a natural crustal cross section of about 10 km (increasing from North to South) in the Dharwar Craton, to study the characteristics of paleomagnetism and geochronology in vertical dimension. U-Pb/Pb-Pb dating on baddeleyite gives a crystallisation age of 2216.0 ± 0.9 Ma for long dyke AKLD. Paleomagnetic data from this well dated ~ 2216 Ma dyke swarm in Dharwar Craton are of excellent quality. High coercivity and high blocking temperature components are carried by single domain magnetite and show dual polarity remanence directions. Combined normal and reverse polarity remanences on AKLD and other N-S dykes define the most reliable paleomagnetic pole for ~ 2216 Ma at latitude 36°S and longitude 312°E (A95 = 7°). Though paleomagnetic data is unavailable on other N-S dykes below the Cuddapah basin, high precision geochronology suggest that they are of similar age within errors. Though there is a variation in the crustal depth of Dharwar craton from north to south, consistent Pb-Pb/U-Pb baddeleyite geochronology and paleomagnetic studies along the AKLD established its continuity and preservation along its entire strike length. The virtual geomagnetic poles of these sites confirm a stable remanence and are almost identical to the previously reported paleomagnetic pole and also supported by positive reversal test. Positive paleomagnetic reversal test on these dykes signify that the remanent magnetization is primary and formed during initial cooling of the intrusions. Updated apparent polar wander path of Dharwar craton indicates relatively low drift rate during 2.21–2.08 Ga interval. Magnetogranulometry and SEM studies show that remanent magnetization in this dyke was carried by single domain magnetite residing within silicate minerals.

1. Introduction

The Earth's history has recorded numerous periods of accentuated mafic magmatism in the form of distinct dyke swarms. Several giant dyke swarms are admirable time markers and generally represent deeply eroded large igneous provinces (LIPs) sections. Precise age determination of such large dyke swarms not only enables establishing the incidence and extent of major magmatic events through time in a given region, but also provides geodynamic linkages to regional scale uplift, continental rifting and break-up and climatic conditions. In conjunction with paleomagnetic data, reliable age determinations of dykes are important for Paleo-continental reconstructions. The Paleoproterozoic of Dharwar Craton witnessed and episodic thermal conditions, especially extensional tectonic regimes that are manifest in multiple swarms of mafic dykes and development of large sedimentary basins.

The Dharwar Craton exposes a tilted Archean crustal cross section of about 10 km, exposing rocks of increasing metamorphic grade, greenschist and amphibolite to granulite facies from north to south (described in detail in the following section). Recently, the occurrence of a 450 km long N-S striking Paleoproterozoic (2215.9 ± 0.3 Ma, Pb-Pb baddeleyite age, Kumar et al., 2014) dyke (Fig. 1) was reported (Kumar et al., 2012b and named as Andhra-Karnataka Long Dyke (AKLD) after the two states in which it occurs) in this region straddling across the different isograds.

This paper focuses on paleomagnetism and geochronological studies on the AKLD along its entire strike length, to study magnetic mineral characteristics at different crustal depths. New paleomagnetic and Pb-Pb baddeleyite age determinations were carried out on the northern extent of the AKLD, in addition to the published data from southern extent on same dyke, to establish the continuity of this dyke intrusion and also to evaluate the effect of regional tectonic activity on the dyke subsequent to its emplacement.

* Corresponding author at: TSMS Jr College, Baswapur, Sangareddy Ditrict, Telangana 502293, India.
E-mail address: enraju@gmail.com (E. Nagaraju).

https://doi.org/10.1016/j.pepi.2017.11.006
Received 13 June 2017; Received in revised form 10 November 2017; Accepted 14 November 2017
Available online 15 November 2017
0031-9201/ © 2017 Published by Elsevier B.V.

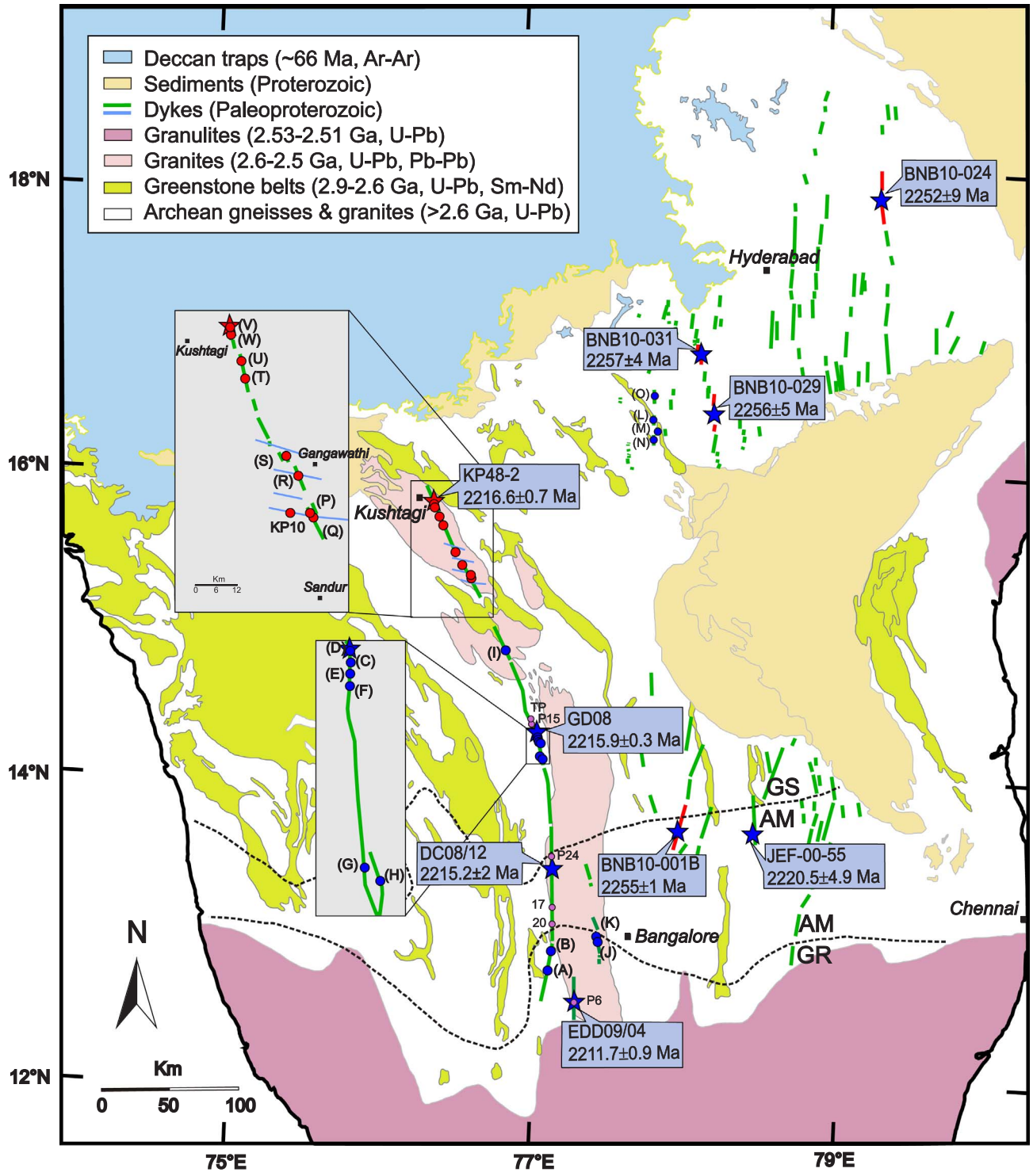


Fig. 1. (a) Simplified geological map of Dharwar craton showing the distribution of N-S dykes and locations of sampled sites. The red filled circles represent sites of paleomagnetic studies in the present study, blue filled circles are sites on which paleomagnetic data were published by Kumar et al. (2012b) and pink circles represent published paleomagnetic sites from Belica et al. (2014) and Piispa et al. (2011). Red and blue stars indicate the site locations of the samples used for Pb-Pb/U-Pb age determination on N-S dykes in the present study and published respectively. GS: Green Schist, AM: Amphibolites, GR: Granulite facies. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2. Regional geology and sampling

Dharwar Craton of south India is the largest and one of the most extensively studied Archean craton in the Indian Peninsular shield. It consists of the southern granulite, Eastern Ghats granulite terrains and the low-grade northern block, which is further classified into two parts viz. Western Dharwar Craton (WDC) and Eastern Dharwar Craton (EDC) on the basis of divergent geological and tectonic characteristics (Swami Nath and Ramakrishnan, 1981; Naqvi and Rogers, 1987; Chardon et al., 2008; Meert et al., 2010). An arcuate N-S trending shear zone that forms the eastern boundary of the Chitradurga greenstone belt is considered to separate western and eastern parts of the Dharwar craton (Chadwick et al., 1992, 2000). Amalgamation of the western and eastern Dharwar cratonic blocks is interpreted to have taken place at ~2.5 Ga (Chadwick et al., 1997).

Based on the progression in metamorphic grade from granulite facies in the south to greenschist facies in the northern part, it has been shown that the Dharwar Craton exposes a natural crustal cross section of a Neoarchean crust (Rollinson et al., 1981; Harris and Jayaram, 1982; Moyen et al., 2003). P-T estimates on the Peninsular Gneisses (Janardhan et al., 1982; Hansen et al., 1984; Gopalakrishna et al., 1986; Stähle et al., 1987; Sen and Bhattacharya, 1990), metapelites (Srinivasan and Tareen, 1972; Rollinson et al., 1981; Harris and Jayaram, 1982) and or mafic rocks (Raase et al., 1986) of the greenstone belts, show P-T conditions vary from nearly 8 kbar and 800 °C (granulite facies) at 12° latitude, to 3.5 kbar and 500 °C (greenschist facies) at 15°N (Fig. 1). The present horizontal distance between the deepest and shallowest levels of the crust is about 350 km. The dip of the present erosion surface relative to the Neoarchean paleo-depth is about 2° (equivalent to a vertical distance of 10–13 km).

As in many Archean blocks, dyke swarms are widespread in the Dharwar craton and have been described in detail earlier (Halls, 1982; Murthy et al., 1987; Halls et al., 2007; French and Heaman, 2010). These dykes range in age from Paleoproterozoic (French and Heaman, 2010; kumar et al., 2012a, 2012b, 2014, 2015; Belica et al., 2014) to Late Cretaceous (Kumar et al., 2001). Of these, the most dominant is the E-W to ENE-WSW trending giant radiating dyke swarm emplaced at ~2367 Ma with an aerial extent of nearly the entire eastern Dharwar craton (Kumar et al., 2012a; Halls et al., 2007). Other dyke swarms include a N-S oriented swarm at 2220.5 ± 4.9 Ma (this could be part of the AKLD swarm), a NW-SE striking swarm at 2209.3 ± 3.8 Ma, a radial WNW-ESE to NW-SE swarm at 2180.8 ± 0.9–2176.5 ± 3.7 Ma (French and Heaman, 2010) and a NE-NW radiating Cuddapah dyke swarm at 2081.8 ± 1.1 Ma (Kumar et al., 2015).

The AKLD sampled in this study is one of the largest single dykes in the Dharwar craton reported so far, it runs nearly parallel to the eastern margin of the Chitradurga schist belt to its east, along the western edge of the eastern Dharwar craton. It can be followed for well over 450 km. In its southern most part, east of the Kunigal schist belt, the dyke intrudes mid-Archaean amphibolite grade gneisses, while along its middle and northern exposure it cuts late-Archaean K-rich granites, low-grade gneisses and schists such as the Sandur schist belt (Fig. 1). The strike of the dyke is generally NNE in the southern and N-S in the middle parts, but swings ~25° NNW in its northern half, thus broadly following the regional structural grain. Thickness of the dyke varies between 40 and 130 m. The dyke preserves a vertical dip along the entire strike length sampled. The dyke is medium to coarse grained and has sharp contacts with the country rock.

A total of 48 (between 5 and 8 oriented block samples from each site) samples were collected for paleomagnetic studies at 8 locations, in addition to the previously reported 9 sites (A to I) in Kumar et al. (2012b) along the strike length of AKLD. Another site (KP 10; 5 samples) is also sampled on E-W dyke near AKLD which is dated at 1894 Ma (Halls et al., 2007). These samples were precisely oriented with magnetic compass. Site names along with their locations are given in Fig. 1 and Table 2. AKLD sites A and B are located on the southern most exposure and are in the granulite facies country rocks, sites C to I, and P to W are located in the central and northern most regions of the dyke exposure respectively, but are all in the green schist facies rocks. Sites P, Q, R and S were collected at the contacts of younger E-W striking dykes for bake contact studies. Though sites P, Q and S are considerable distance (more than 250 m) from E-W dykes, site R is very close (about 20–30 m) to one of the E-W dyke. One coarse grained sample from the northern most site V (sample KP 48-2) was selected for Pb-Pb baddeleyite geochronology. Our sampling was confined mostly to fresh outcrops at road cuts and quarries. Petrographic description and geochemical characterization of the AKLD was reported earlier by Kumar et al., (2012b). The major mineral constituents of this dyke include plagioclase, augite and occasionally hypersthene, with minor amounts of magnetic minerals (modal abundance 4–5%). Most samples exhibit poikilitic texture, though in a few sections porphyritic texture was also observed. Though the degree of metamorphism in the country rock increases from north to south the extent of alteration or clouding in feldspar (Halls and Zhang, 1995) does not vary along strike. AKLD is basaltic tholeiite in composition with minor variation along strike (Kumar et al., 2012b).

3. Experimental procedures

3.1. Paleomagnetic and rock magnetic studies

Sample susceptibility variation with temperature for determining Curie temperature was conducted on Bartington Instrument model MS-2. A total of 76 specimens from eight of the 17 sites (data from the other nine sites were reported earlier, Kumar et al., 2012b) were subjected to standard demagnetization procedures including incremental alternating field (AF) and heat treatment (TH) were performed using Molspin Alternating Field demagnetizer and MMTD 80 oven (Magnetic Measurements limited) respectively. After each incremental demagnetization step, the natural remanent magnetization (NRM) was measured using JR-6 spinner magnetometer (Agico).

3.2. Imaging of magnetite grains

Structure and morphology of magnetite grains were studied using Hitachi, S-3400N Scanning Electron Microscope (SEM) and their Back-Scattered Electron (BSE) images. Spot analyses were also performed using Energy-Dispersive X-ray Spectroscopy (EDS) to know the chemical composition of the different magnetite phases observed.

3.3. Pb-Pb baddeleyite geochronology

Baddeleyite age determination was done using the TE-TIMS (Thermal Extraction – Thermal Ionization Mass Spectrometry) following the procedure described by Kumar et al., (2014). Baddeleyite grains were separated from a 200 g sample using a Wilfley table, followed by handpicking under a binocular microscope. Best quality (mineral inclusions, fractures and alteration free) baddeleyite crystals were cleaned and loaded on out gassed zone refined rhenium filaments (0.030 in. wide by 0.001 in. thick) and annealing at 1250 °C in a vacuum chamber at $\sim 1 * 10^{-7}$ mbar for 30 min to expel disturbed Pb from altered domains. Grains were subsequently embedded in silica glass on the same Re filament by fusing the sample with silica gel and phosphoric acid mixture at 1200 °C for ~10 min. in a vacuum chamber. Pb isotopic analyses were performed on a TRITON^{PLUS} Thermal Ionization Mass Spectrometer at the CSIR-National Geophysical Research Institute, Hyderabad. The three high mass Faraday detectors (H1 to H3) connected to 10^{11} ohm resistors were used to measure ^{206}Pb, ^{207}Pb and ^{208}Pb, in static multi-collector mode with virtual amplifier rotation. The ^{204}Pb signal was very small and therefore was collected simultaneously by a secondary electron multiplier – ion counting system on the axial channel. Isotopic ratios were corrected for instrumental mass fractionation of 0.18%/atomic mass unit.

The ^{207}Pb/^{206}Pb ages for each block were calculated from the determined ^{207}Pb/^{206}Pb and ^{204}Pb/^{206}Pb ratios with corrections for common Pb according to the model proposed by Stacey and Kramers (1975). Ages are determined from weighted means of fraction averages using accepted decay constants. Details regarding data acquisition and data reduction are given in Kumar et al., (2014).

4. Results

4.1. Thermomagnetic and SEM analysis

Magnetic mineralogy as determined from thermomagnetic curves of low field susceptibility with reversible thermomagnetic curves derived from samples KP 38 (site T) and KP 56 (site W) shows a Curie

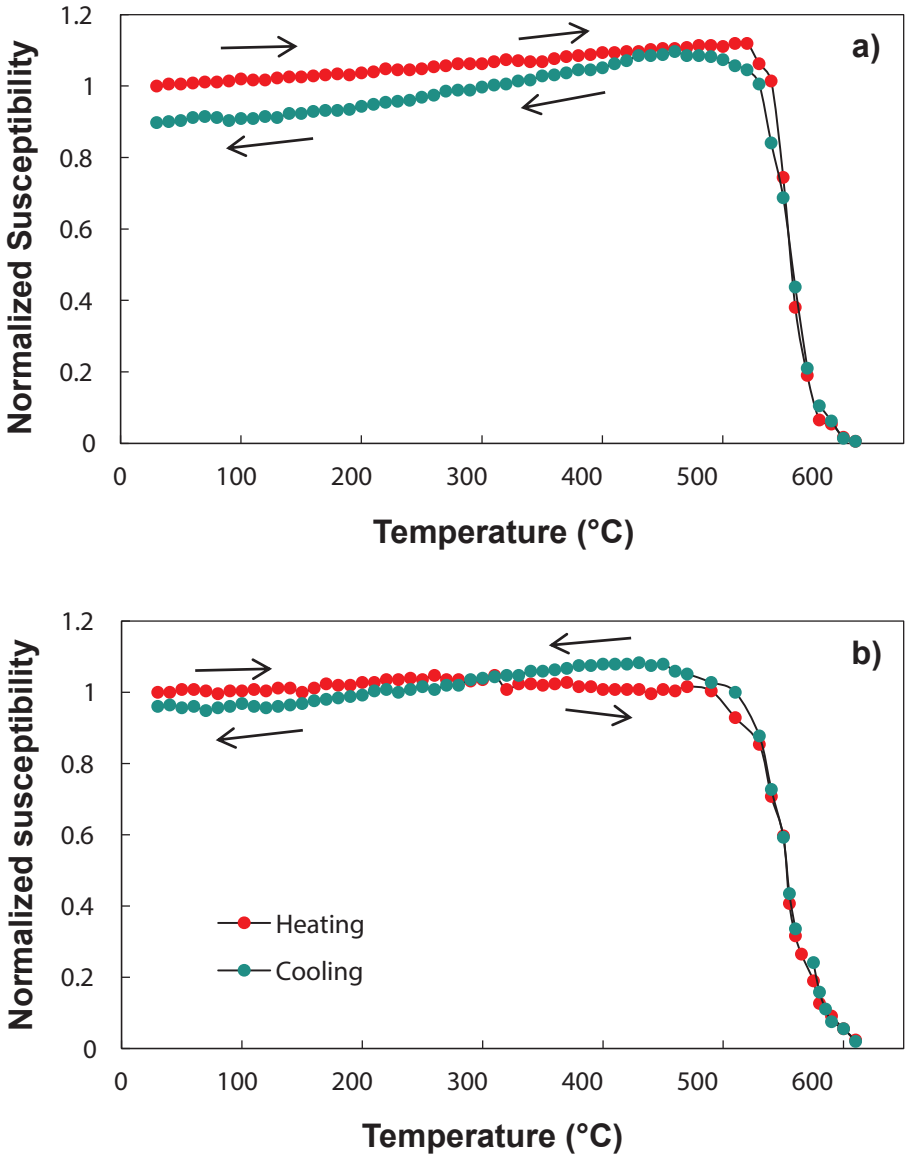


Fig. 2. Typical examples of magnetic susceptibility versus temperature dependencies.

temperature close to 580 °C indicates that the dominant magnetic carrier is Ti-poor magnetite (Fig. 2).

However, SEM images (Fig. 3a and b) clearly show two types of magnetic minerals. Fine grained needle like single domain (1 × 5 μm) magnetite located within silicate minerals such as pyroxene and plagioclase. Though this constitutes a very small modal percentage of the rock it could be the main carrier of remanent magnetization in this dyke. Medium grained multidomain (~50 μm) Ti-poor magnetite occurs more ubiquitously (4–5%) as interstitial grains in these samples.

4.2. Geochronology of AKLD

Four baddeleyite fractions were analysed from sample KP 48-2 (site V). Each fraction average contains at least 19 blocks. The grand weighted mean Pb-Pb TE-TIMS age for the four fractions from this sample is 2216.6 ± 0.7 Ma (MSWD = 0.57, after adding fractionation uncertainty of 0.055%, as explained in Kumar et al. (2014)). Results are given in Table 1 and plotted in Fig. 4. This age is identical within error with previously reported weighted mean ^{207}Pb–^{206}Pb baddeleyite TE-

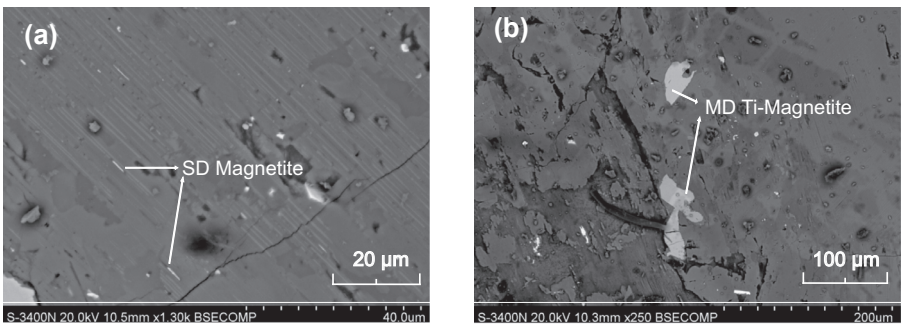


Fig. 3. Scanning electron microscope (SEM) back-scattered electron images of (a) single domain magnetite within a pyroxene grain and (b) interstitial multidomain Ti-poor magnetite.

Table 1
TE-TIMS Pb isotopic data on baddeleyite fractions from KP 48-2 sample (AKLD).

Sample No.	Samp. wt.	No. bloc.	$^{206}Pb/^{204}Pb$ (m)	Abs. error	$^{207}Pb/^{206}Pb$ (m)	$^{207}Pb/^{206}Pb$ (c)	Pb-Pb Age (Ma)
KP 48-2							
KP 48-2-1	6.1	23	23201	484	0.139704 ± 015	0.139138 ± 019	2216.5 ± 1.4
KP 48-2-2	4.9	19	6755	281	0.141142 ± 094	0.139164 ± 012	2216.9 ± 1.7
KP 48-2-3	5.1	19	3431	117	0.143061 ± 135	0.139184 ± 020	2217.1 ± 1.5
KP 48-2-4	5.4	25	8477	145	0.140630 ± 014	0.139083 ± 017	2215.8 ± 1.5
Grand weighted mean after adding fractionation uncertainty of 0.055%.							
Age: 2216.6 ± 0.7 Ma (MSWD = 0.57)							

Sample wt. is in micrograms. $^{206}Pb/^{204}Pb$ (m) and $^{207}Pb/^{206}Pb$ (m) are measured values (fraction means with standard error) and $^{207}Pb/^{206}Pb$ (c) is the corrected value (wtd. means and errors). Uncertainties are $2\sigma_m$ and refer to the least significant digits.

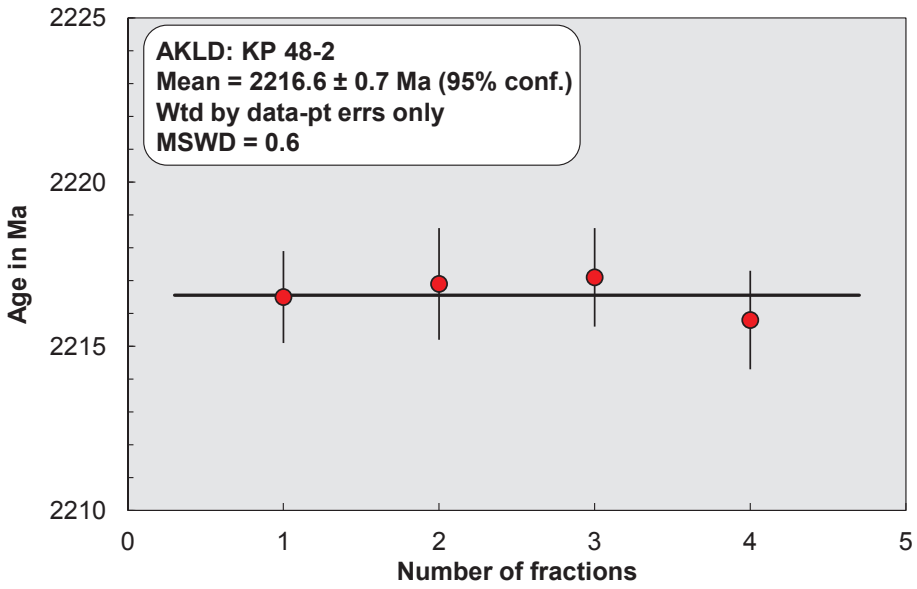


Fig. 4. TE-TIMS weighted mean $^{207}Pb/^{206}Pb$ ages on four baddeleyite fractions from sample KP 48-2 of AKLD. Error bars represent 95% confidence limits of measurements.

TIMS age of 2215.9 ± 0.3 Ma (Kumar et al., 2014) for sample GD-08 from site D (Fig. 1) and a conventional U-Pb age of 2215.0 ± 2.0 Ma for the DC08/12 sample (Srivastava et al., 2011), which is about 100 km south of GD-08 sites (Fig. 1). A weighted mean of the three age determinations on the AKLD gives a grand mean age of 2216.0 ± 0.9 Ma which we consider as the best age estimate for the emplacement this dyke. The three overlapping ages along the entire 450 km length suggests this dyke was emplaced in a very short duration.

4.3. Paleomagnetism

Paleomagnetic results on AKLD from nine sites A, B, C, D, E, F, G, H and I, all located (Fig. 1) on the southern and central parts of the dyke were reported earlier (Kumar et al., 2012b). The mean D_m, I_m and paleomagnetic pole reported for these sites were 228°/−61° and 40°S/304°E respectively. The new paleomagnetic data presented here on AKLD are from sites P, Q, R, S, T, U, V and W. At least two specimens from each sample were subjected to detail stepwise AF and/or thermal demagnetization, to a maximum of 150 mT or 600 °C, in order to separate magnetic components according to coercive force and/or blocking temperature. Paleomagnetic results are presented in Table 2 and plotted in Figs. 5 and 6. Paleomagnetic directional data were deduced using Kirschvink's (1980) principle component analysis procedure in combination with Zijderveld (1967) diagrams. Acceptance criteria for linear segments on vector diagrams were the number of consequent demagnetization points (≥5) and the maximum angle of deviation ≤10° (Kirschvink, 1980). Though all 8 sites (P to W) are on the single dyke, sites T, U, V and W shows a characteristic remanent magnetization (ChRM) direction similar to the southern sites reported earlier (Fig. 6a). Where as, sites P, Q, R and S, as mentioned in the preceding section that were collected at the contacts within half the thickness of younger E-W dykes (which vary between 10 and 25 m thick), record a direction that is identical to the ChRM of the younger 1890 Ma dykes and Cuddapah traps (Kumar and Bhalla, 1983; Clark, 1982; Belica et al., 2014) in this region (Table 2 and Fig. 6b). The site R gives a site mean characteristic direction of D_m/I_m = 295°/−31° and is close to the direction obtained from site KP 10 (E-W dyke), which is dated as 1894 Ma (Halls et al., 2007) and gives a mean D_m/I_m as 314°/−12°. These two sites directions are within the errors of previously reported NW dykes (Kumar and Bhalla, 1983) and Cuddapah Traps (Clark, 1982). Directions obtained from baked sites P, Q and S are unknown. Apart from these sites on AKLD, Piispa et al. (2011) and Belica et al. (2014) reported similar directions on five more sites (sites P15, P24 and TP, 17 and 20 respectively) on AKLD along with a single but most significant reverse magnetic direction on site P6 from another dyke (Table 2). The reverse directions obtained on two sites (sites P15 and P6) are approximately antipodal to that of normal directions. Integrating normal and reverse directions share a common mean direction at 36°S and 312°E (Fig. 6a). The reversal test was conducted to test antipodality of the means and resulted in a classification of R_c (observed λ = 6.1, Critical λ = 19.2; McFadden and McElhinny 1990) for this dyke swarm.

5. Discussion

Dyke remanences may also be considered as primary if reversals in the paleofield are recorded by different dykes in a swarm, because this

Table 2
Results of paleomagnetic measurements on 2.21 Ga N-S Dharwar dyke swarm.

Site	Lat. (°N)	Long. (°E)	S°	N	D_m	I_m	k	α_{95}	p lat. (°N)	p long. (°E)	A_{95}	Ref.
Component '2216 Ma'												
T	15.662	76.342	350	5	248	−68	60	10	−26	297	15	This study
U	15.705	76.331	338	9	272	−63	30	10	−10	303	13	This study
V	15.777	76.301	358	8	215	−53	14	16	−54	311	18	This study
W	15.773	76.301	350	6	295	−49	67	8	−13	310	9	This study
A	12.767	77.056	14	7	243	−55	39	10	−30	314	12	1
B	12.851	77.069	357	6	245	−63	100	7	−27	304	9	1
C	14.216	76.977	1	10	204	−60	190	4	−56	291	4	1
D	14.218	76.977	359	12	218	−66	60	6	−43	291	8	1
E	14.209	76.977	357	5	220	−55	58	10	−48	310	12	1
F	14.204	76.977	358	7	240	−59	35	10	−32	309	13	1
G	14.117	76.985	350	15	231	−55	83	4	−40	312	5	1
H	14.112	76.993	343	9	236	−63	80	6	−34	303	8	1
I	14.778	76.786	335	7	215	−65	27	12	−47	291	17	1
J	12.974	77.353	354	6	238	−52	123	6	−34	317	7	1
K	12.950	77.378	350	5	253	−59	504	3	−21	309	4	1
TP	14.387	76.916	350	6	230	−57	33	12	−40	310	–	2
P15	14.368	76.907	345	6	38	62	20	16	−47	297	–	3
P24	13.537	77.048	345	9	236	−48	15	14	−36	321	–	3
17	13.183	77.041	345	4	218	−69	243	6	−40	287	–	2
20	13.061	77.037	358	8	282	−47	39	9	−4	317	–	2
P6	12.498	77.234	0	5	85	67	10	25	−13	299	–	3
Mean (N + R)	**–**	**–**	**–**	**21**	**240**	**−61**	**34**	**6**	**−36**	**312**	**7**	
Component '1894 Ma'												
KP 10 #	15.301	76.481	90	10	290	−29	61	6	14	326	5	This study
R	15.404	76.487	338	11	295	−31	41	7	18	323	6	This study
NW dyke	14.400	77.700	–	9	320	−34	68	13	39	309	–	4
Cuddapah Traps	14.300	78.200	–	15	299	−6	18	16	27	337	–	5
1.9 Mean	15.707	79.050	–	29	129	9	18	7	36	331	–	2
Mean	**14.851**	**77.217**	**–**	**5**	**303**	**−22**	**23**	**16**	**27**	**326**	**14**	This study
Other component												
P	15.292	76.529	341	4	331	2	50	13	−58	323	10	This study
Q	15.303	76.521	344	7	326	−5	64	8	−52	322	6	This study
S	15.456	76.452	337	10	323	6	148	4	−51	332	3	This study

Note: Lat. & Long. = latitude and longitude; S^o = is strike of the dyke at the respective sites. N = number of samples studied from each site; D_m = mean declination; I_m = mean inclination; k = precision parameter; α_{95} and A_{95} = the radius of the 95% circle of confidence about the mean magnetization direction for site and pole respectively; p lat. and p long. = paleocoordinates of the pole; all statistical parameters are based on sample means. Angles (D_m, I_m, α_{95} and A_{95}) are in degrees. # = site dated at 1894 Ma (Halls et al., 2007). Ref.: 1 = Kumar et al. (2012a,b); 2 = Belica et al. (2014); 3 = Piispa et al. (2011); 4 = Kumar and Bhalla, 1983; 5 = Clark (1982).

pattern would not result from metamorphic cooling. However, to conduct this test sampling must be carried out at sites along dykes of different magnetic polarity in a given area to establish that polarity is consistent along each dyke. Paleomagnetic data from AKLD and other N-S dykes are of excellent quality, and both normal and reversed polarities recorded from dyke sets share a common mean at the 95% confidence level. The combined direction is indistinguishable from the dominant AKLD direction of magnetization reported in earlier studies of Dharwar craton. Though only two sites represent reverse magnetic directions, reversal test on limited sites provides positive and places the combined direction in 'C' category. Both the AKLD and other N-S dyke remanences are interpreted to be primary on the basis of positive paleomagnetic reversal test. Given the positive reversal test, component isolated from all dykes in this study were inverted to the same polarity before calculating mean paleomagnetic pole (Table 2).

Due to the variation in the crustal depth of about 10 km from north to south due to the tilted cross section of Dharwar craton, this could result in the huge variation in temperature (about 200 °C–250 °C; as going into the depth of crust, increase in temperature is of about 20–25 °C) between two crustal depths at the time of rock formation. This indicates that dyke formed at shallow end has cooled faster than that of deeper end, which should result in preserving magnetic memory during considerable time interval. However, either the remanence magnetic directions or baddeleyite geochronological results along the entire dyke length do not reflect this behaviour and confirms that AKLD has emplaced in a very short time interval of less than 1 Ma.

To use the paleomagnetic data obtained from 2216 Ma dyke swarm in this study as 'key pole', it is mandatory to ascertain if the calculated Virtual Geomagnetic Pole (VGP) has averaged out Secular Variation (SV). The calculated VGP scatter value (Cox, 1970) for the sites studied on this dyke swarm is 16.8° +2.5°/−5.1° and is similar to the observed VGP angular dispersion value of about 16° (Johnson et al., 2008 and McElhinny and McFadden, 1997) at the same latitude (Dharwar N-S dyke swarm paleolatitude of 45°S), during the last 0–5 Ma period suggesting that SV was adequately averaged out.

The available Paleoproterozoic paleomagnetic data for the Dharwar craton reported by recent studies were recalculated to obtain paleomagnetic pole positions corresponding to normal polarity (Table 3). The paleopoles obtained from precisely dated rocks for the Paleoproterozoic period form four clusters" (i.e. 2.36–2.37 Ga, 2.21–2.22 Ga, ~2.08 Ga and ~1.89 Ga). Based on the above well defined clusters of pole positions, an updated APWP for Dharwar craton during Paleoproterozoic time is reconstructed (Fig. 7). Drift rates calculated from the APW path shows that Dharwar craton has drifted at a rate of 5.1 cm/yr during 2.37–2.21 Ga, 3.1 cm/yr during 2.21–2.08 Ga and 6.3 cm/yr during 2.08–1.89 Ga. This shows that the Dharwar craton has drifted relatively very slow during 2.21–2.08 Ga, which might be due to any one of the major tectonic activity related to that period.

The two Pb-Pb TE-TIMS (Kumar et al., 2014 and the new age reported here) and the U-Pb baddeleyite (Srivastava et al., 2011) ages (grand mean age of 2216.0 ± 0.9 Ma) on the AKLD are identical within error and suggest its emplacement along its entire strike length to be with in a geologically short span of less than ~1 Ma. The N-S striking Kandlamadugu dyke with a similar emplacement age of

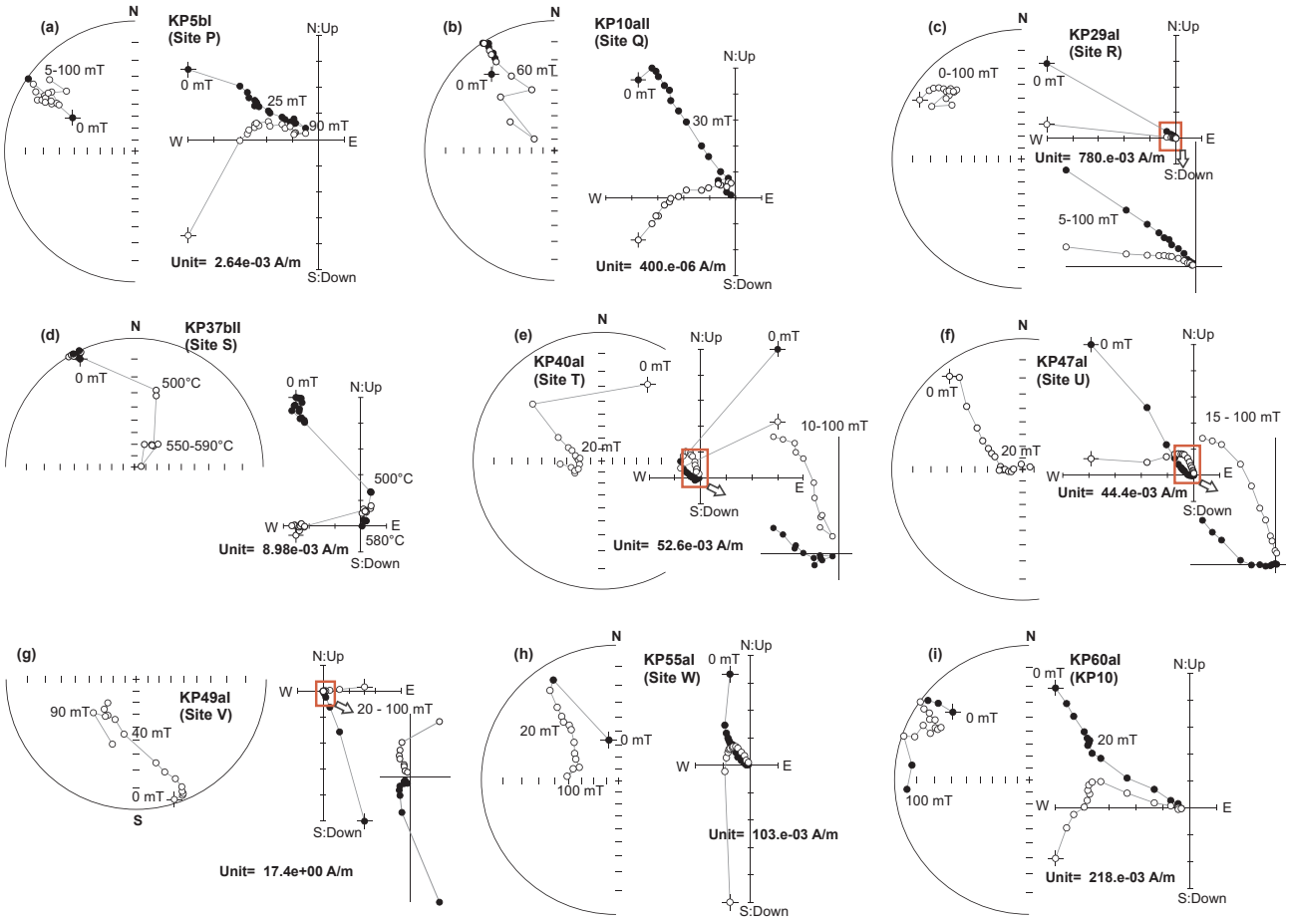


Fig. 5. Zijderveld diagram and equal area stereonet projections showing characteristic behavior of natural remanence to AF (diagrams a–c and e–i) and thermal demagnetization (diagram d) for samples from representative sites. Thermal measurements are in °C and AF measurements are in millitesla (mT). Open/closed circles in the stereo plots represent upward/downward directed vectors and open/closed circles in the Zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0 plotting and analysis program (Chadima and Hrouda, 2006).

2220.5 ± 4.9 Ma (French and Heaman, 2010) could also be a part of the AKLD swarm. N-S striking dykes from the north of Cuddapah basin are all different in their recently reported U-Pb/ Pb-Pb baddeleyite ages. The precise Pb-Pb geochronology on BNB10-029 (2256 ± 4 Ma), BNB10-031 (2257 ± 4 Ma), BNB10-024 (2252 ± 9 Ma) and BNB10-001B (2255 ± 1 Ma) provides unambiguous evidence that all N-S dykes in this craton are not a part of single dyke swarm (Demirer, 2012). Keeping this in the view and slightly steeper inclinations from previously reported four sites on dyke iii (Kumar et al., 2012b) were not included in 2216 Ma dyke swarm. Dyke iii could be a part of ~2252 Ma dyke swarm (Sites O, L, M, and N in Fig. 1).

Mafic magmatism at ~2216 Ma has been reported from several continents including North America, Greenland, Europe, Africa and Australia. The impact of this event was most pronounced in North America, within the Superior province and includes the Senneterre dykes that were emplaced at 2216 +8/−4 Ma (Buchan et al., 1998), Nipissing sills 2217 ± 4 Ma (Noble and Lightfoot, 1992) and the Kogaluk bay dyke swarm 2212 ± 3 Ma (Maurice et al., 2009). Such wide spread distribution of this event suggests it had a global impact.

Implications of the paleomagnetic data from sites A through I on the AKLD were previously reported (Kumar et al., 2012b) and a Paleoproterozoic reconstruction of the Dharwar, Slave, Superior and Rae cratons attempted. Since the VGP obtained for new sites including normal and reverse directions are similar to earlier reported sites, we do not want to repeat that discussion here. However, the additional four sites from the northern half of the AKLD enable us to understand the apparent 25° north-westerly swing in the dyke trend, whether this was a primary emplacement feature or caused due to subsequent tectonic rotation of the northern part with respect to the southern, similar to the curvature observed in the regional structural grain of the Archean basement. To address this problem, site mean declination vs. strike along the dyke were plotted in Fig. 8. No clear correlation is seen between dyke trend and the paleomagnetic directions. The observed scatter in the declination may not reflect any tectonic rotation. Therefore, the precise age determination and reliable paleomagnetic data with primary remanent magnetic vectors recorded by single domain high blocking temperature magnetic minerals, widely distributed (~450 km) site locations averaging out secular variation and a positive reversal test satisfies the basic reliability criteria suggested by Buchan et al., (2000) to qualify as a key pole. This will therefore be the Paleoproterozoic paleomagnetic 'key pole' for 2216 Ma swarm from the Dharwar craton.

6. Conclusion

Consistent and precise Pb-Pb and U-Pb baddeleyite age determinations along the dyke establish AKLD as a single dyke entity running for over 450 km and exposing a north-south cross section of about 10 km at crustal depths of about 15–25 km deep.

Paleomagnetic data along the AKLD on 17 sites present consistent remanent magnetic directions with a positive reversal test, is extremely good evidence to demonstrate that paleomagnetic data from different mid crustal levels are not obliterated by the large compressive stresses

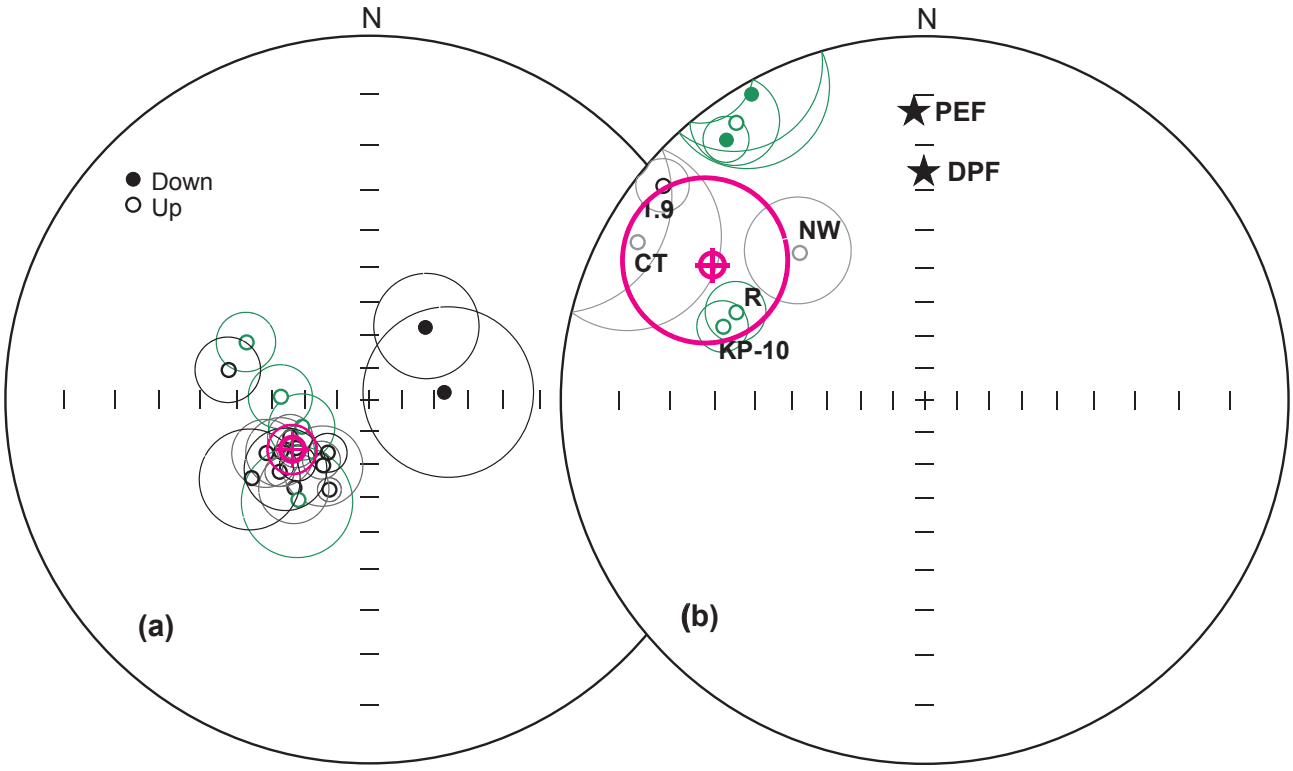


Fig. 6. Stereo plots showing paleomagnetic data of all the 17 AKLD sites. (a) Primary site mean directions with ovals of 95% confidence of 13 AKLD sites. Published and present study are in black and green colour respectively. Pinck circle represents grand mean of AKLD sites. (b) Baked directions (from sites P, Q and S) with ovals of 95% confidence and direction obtained from site R and KP 10 are in green colour. Similar published directions from Clark (1982), Kumar and Bhalla (1983) and Belica et al. (2014) are in grey colour. Pink circle represents combined mean of published ~1890 Ma directions. Black stars represent DPF (Dipole field) and PEF (Present Earth's field) direction based on the 1995 IGRF. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Paleomagnetic data of the dykes reported from Dharwar craton used to reconstruct an updated APWP for Paleoproterozoic period.

Rock unit	Age in Ma	Dec	Inc	Pole Lat. (°N)	Pole Long. (°E)	Refs.
Giant radiating dyke swarm (2.36–2.37 Ga)	2365 ± 1.5	97	−79	15	57	Rameshbabu et al. (2017) and Kumar et al. (2012a)
AKLD swarm (2.21–2.22 Ga)	2215.9 ± 0.3	240	−61	36	132	Kumar et al. (2012b) & Present study
Cuddapah radiating dyke swarm (~2.08 Ga)	2081.8 ± 1.1	51	1	38	180	Kumar et al. (2015)
~1.88 Ga swarm	1885.4 ± 3.1	303	−22	27	326	French et al. (2008), Present Study

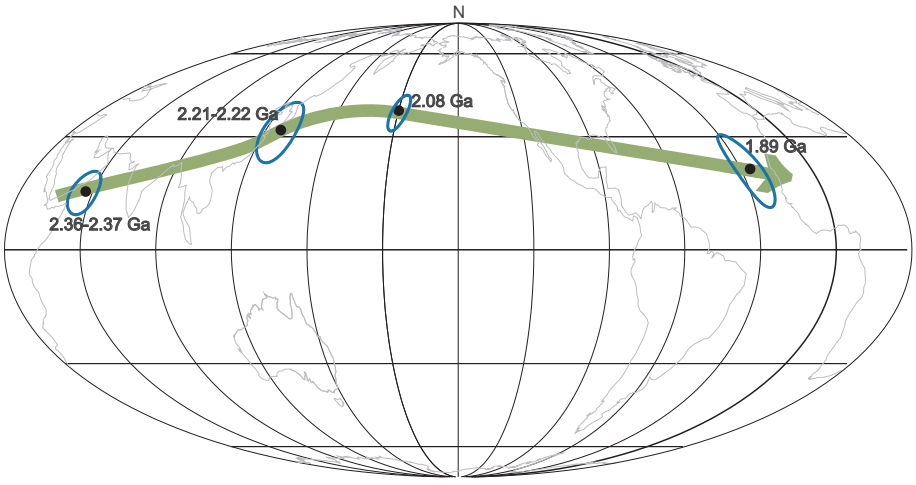


Fig. 7. Refined Apparent polar wander path (APWP) drawn for Paleoproterozoic period for Dharwar craton. Details of reference poles are given in Table 3.

exerted by the thick (> 15 km) overlying upper crust.

Though the AKLD strike swings NNW at its northern end, the observed declinations from this part of the dyke do not show a sympathetic rotation to support a post emplacement tectonic activity of the region. Together with precise age determination, consistent paleomagnetic data and a positive reversal test, 2216 Ma dyke swarm will therefore be the Paleoproterozoic paleomagnetic 'key pole' from the Dharwar craton.

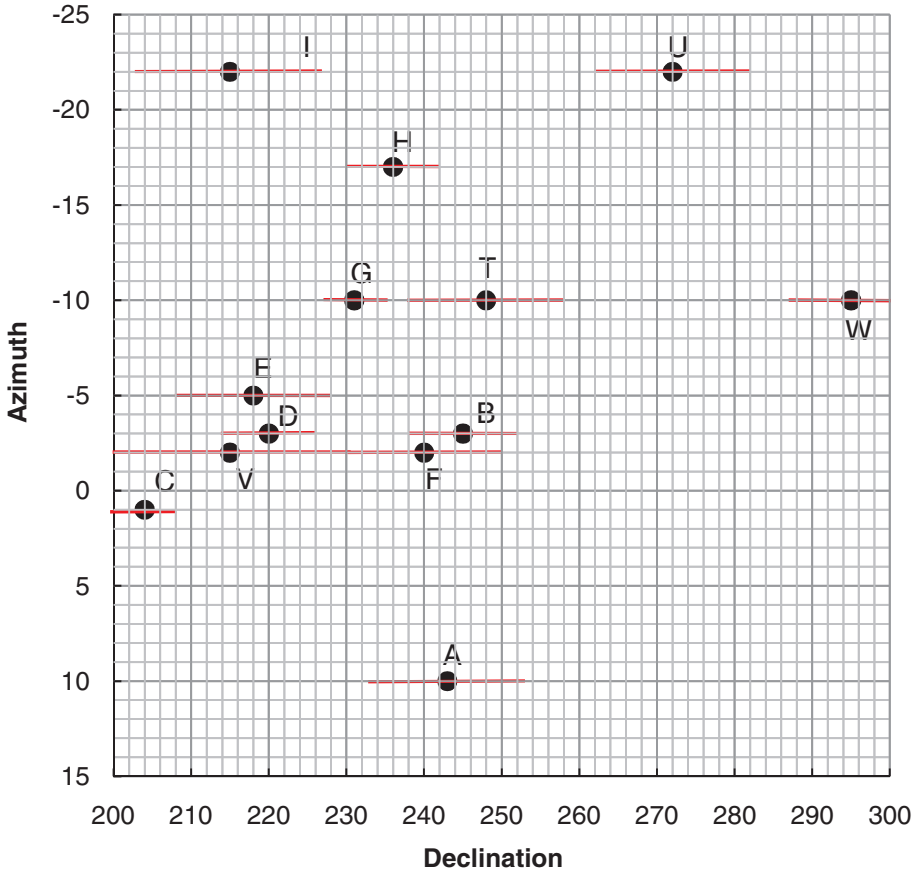


Fig. 8. Plot showing site paleomagnetic declination vs. dyke strike (azimuth).

Acknowledgements

We thank the Director, National Geophysical Research Institute, Hyderabad for his encouragement to publish this work. We thank Dr. N. Ramesh Babu for paleomagnetic measurements. This work was supported by the NGRI MLP 6514 and INDEX project funds.

References

Belica, M.E., Piispa, E.J., Meert, J.G., Pesonen, L.J., Plado, J., Pandit, M.K., Kamenov, G.D., Celestino, M., 2014. Paleoproterozoic mafic dyke swarms from the Dharwar Craton; paleomagnetic poles for India from 2.37–1.88 Ga and rethinking the Columbia supercontinent. Precambrian Res. 244, 100–122.

Buchan, K.L., Mertanen, S., Park, R.G., Pesonen, L.J., Elming, S.-Å., Abrahamsen, N., Bylund, G., 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key palaeomagnetic poles. Tectonophysics 319 (3), 167–198.

Buchan, K.L., Mortensen, J.K., Card, K.D., Percival, J.A., 1998. Palaeomagnetism and U-Pb geochronology of diabase dyke swarms of Minto block, Superior Province, Quebec, Canada. Can. J. Earth Sci. 35, 1054–1069.

Chadima, M., Hrouda, F., 2006. Remasoft 3.0 user-friendly data browser and analyzer. Travaux Geophys. XXVII, 20–21.

Chadwick, B., Vasudev, V.N., Hegde, G.V., 1997. The Dharwar craton, southern India, and its Late Archean plate tectonic setting: current interpretations and controversies. Ind. Acad. Sci. (Earth and Planet. Sci.) Proc. 106 (4), 1–10.

Chadwick, B., Vasudev, V.N., Hegde, G.V., 2000. The Dharwar craton, southern India, interpreted as the result of Late Archean oblique convergence. Precambrian Res. 99, 91–101.

Chadwick, B., Vasudev, V.N., Krishna Rao, B., Hegde, G.V., 1992. The Dharwar Supergroup: basin development and implications for late Archean tectonic setting in western Karnataka, Southern India. In: Glover, J.E., Ho, S.E. (Eds.), The Archean: Terrains, Processes and Metallogeny. Univ. West. Aus. Publ. 22, pp. 3–15.

Chardon, D., Jayananda, M., Chetty, T.R.K., Peucat, J.J., 2008. Precambrian continental strain and shear zone patterns: South Indian case. J. Geophys. Res. 113, B08402. http://dx.doi.org/10.1029/2007JB005299.

Clark, D.A., 1982. Preliminary Paleomagnetic Results From the Cuddapah Traps of Andhra Pradesh, Monograph-2, On Evolution of the intracratonic Cuddapah Basin. HPG, Hyderabad, India, pp. 47–51.

Cox, A.V., 1970. Latitude dependence of the angular dispersion of the geomagnetic field. Geophys. J.R. Astron. Soc. 20, 253–269.

Demirer, K., 2012. (Dissertations in Geology at Lund University, Master's thesis) In: U-Pb Baddeleyite Ages From Mafic Dyke Swarms in Dharwar Craton, India – Links to an Ancient Supercontinent, pp. 308.

French, J.E., Heaman, L.M., 2010. U-Pb dating of Paleoproterozoic mafic dyke swarms of the south Indian Shield: implications for paleocontinental reconstructions and identifying ancient mantle plume events. Precambrian Res. 183, 416–441.

Gopalakrishna, D., Hansen, E.C., Janaradhan, A.S., Newton, R.C., 1986. The southern high grade margin of the Dharwar craton. J. Geol. 94, 247–260.

Halls, H.C., 1982. The importance and potential of mafic dyke swarms in studies of geodynamic processes. Geosci. Can. 9, 145–153.

Halls, H.C., Kumar, A., Srinivasan, R., Hamilton, M.A., 2007. Paleomagnetism and U-Pb geochronology of eastern trending dykes in the Dharwar craton, India: feldspar clouding, radiating dyke swarms and position of India at 2.37 Ga. Precambrian Res. 155, 47–68.

Halls, H.C., Zhang, B., 1995. Tectonic implications of clouded feldspar in Proterozoic mafic dyke swarms. Geol. Soc. India Memoir 33, 65–80.

Hansen, E.C., Newton, R.C., Janardhan, A.S., 1984. Pressures, temperatures and metamorphic fluids across and unbroken amphibolites facies to granulite facies transition in southern Karnataka. In: Kroner (Ed.), Archean Geochemistry. Springer-Verlag, Berlin, pp. 161–181.

Harris, N.B.W., Jayaram, S., 1982. Metamorphism of cordierite gneisses from the Bangalore region of the Indian Archaean. Lithos 15, 89–98.

Janardhan, A.S., Newton, R.C., Hansen, E.C., 1982. The transformation of amphibolites facies gneiss to charnockite in southern Karnataka and northern Tamil Nadu, India. Contrib. Miner. Petrol. 79, 132–149.

Johnson, C.L., Constable, C.G., Tauxe, L., Barendregt, R., Brown, L.L., Coe, R.S., Layer, P., Mejia, V., Opdyke, N.D., Singer, B.S., Staudigel, H., Stone, D.B., 2008. Recent investigations of the 0–5 Ma geomagnetic field recorded by lava flows. Geochem. Geophys. Geosystems 9. http://dx.doi.org/10.1029/2007GC001696.

Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Kumar, A., Bhalla, M.S., 1983. Paleomagnetic and igneous activity of the area adjoining the southwestern margin of the Cuddapah basin, India. Geophys. J. R. Astron. Soc. 73, 27–37.

Kumar, A., Hamilton, M.A., Halls, H.C., 2012a. A Paleoproterozoic giant radiating dyke swarm in the Dharwar Craton, southern India. Geochem. Geophys. Geosyst. 13. http://dx.doi.org/10.1029/2011GC003926.

Kumar, A., Nagaraju, E., Besse, J., Bhasker Rao, Y.J., 2012b. New age, geochemical and paleomagnetic data on a 2.21 Ga dyke swarm from southern India: constraints on Paleoproterozoic reconstruction. Precambrian Res. 220–221, 123–138.

Kumar, A., Nagaraju, E., Srinivasa Sarma, D., Davis, D.W., 2014. Precise Pb baddeleyite geochronology by the thermal extraction-thermal ionization mass spectrometry method. Chem. Geol. 372, 72–79.

Kumar, A., Pande, K., Venkatesan, T.R., Bhaskar Rao, Y.J., 2001. The Karnataka Late Cretaceous dyke as products of the Marion hotspot at the Madagascar-India break up event: evidence from 40Ar-39Ar geochronology and geochemistry. Geophys. Res. Lett. 28, 2715–2718.

Kumar, A., Parashuramulu, V., Nagaraju, E., 2015. A 2082 Ma radiating dyke swarm in the Eastern Dharwar Craton, southern India and its implications to Cuddapah basin formation. Precambrian Res. 266, 490–505.

Maurice, C., David, J., O'Neil, J., Francis, D., 2009. Age and tectonic implications of Plaeoproterozoic mafic dyke swarms for the origin of 2.2 Ga enriched lithosphere beneath the Ungava Peninsula, Canada. Precambrian Res. 174, 163–180.

McElhinny, M.W., McFadden, P.L., 1997. Palaeosecular variation over the past 5 Myr based on a new generalized database. Geophys. J. Int. 131, 240–252.

McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism. Geophys. J. Int. 103, 725–729.

Meert, J.G., Pandit, M.K., Pradhan, V.R., Banks, J.C., Sirianni, R., Stroud, M., Newstead, B., Gifford, J., 2010. The Precambrian tectonic evolution of India: a 3.0 billion year odyssey. J. Asian Earth Sci. 39, 483–515.

Moyen, J.F., Nedelec, A., Martin, H., Jayananda, M., 2003. Syntectonic granite emplacement at different structural levels: the Closepet granite, South India. J. Struct. Geol. 25, 611–631.

Murthy, Y.G.K., Babu Rao, V., Guptasarma, D., Rao, J.M., Rao, M.N., Bhattacharjee, S., 1987. Tectonic, petrochemical and geophysical studies of mafic dyke swarms around the Proterozoic Cuddapah basin, south India. In: Halls, H.C., Fehrig, W.E. (Eds.), Mafic Dyke Swarms. Geol. Assoc. Can. Spec. Pap. 34, pp. 303–316.

Naqvi, S.M., Rogers, J.J.W., 1987. Precambrian Geology of India. Oxford Monogr. Geol. Geophys. 6, 233p.

Noble, S.R., Lightfoot, P.C., 1992. U-Pb baddeleyite ages of the Kerns and Triangle Mountain intrusions, Nipissing Diabase, Ontario. Can. J. Earth Sci. 29, 1424–1429.

Piispa, E.J., Smirnov, A.V., Pesonen, L.J., Lingadevaru, M., Ananthamurthy, K.S., Devaraju, T.C., 2011. An Integrated study of Proterozoic dykes, Dharwar craton, Southern India. In: Dyke swarms: Keys for Geodynamic Interpretation. Spinger-Verlag, Berlin/Heidelberg, pp. 33–45.

Rameshbabu, N., Venkateshwarlu, M., Shankar, Ravi, Nagaraju, E., Parashuramulu, V., 2017. New Paleomagnetic results on ~2367 Ma Dharwar giant dyke swarm, Dharwar craton, Southern India: implications for Paleoproterozoic continental reconstruction. J. Earth Syst. Sci (accepted manuscript).

Raase, P., Raith, M., Ackermand, D., Lal, R.K., 1986. Progressive metamorphism of mafic rocks from greenschist to granulite facies in the Dharwar craton of South India. J. Geol. 94, 261–282.

Rollinson, H.R., Windley, B.F., Ramakrishnan, M., 1981. Contrasting high and intermediate pressures of metamorphism in the Archean Sargur schists of southern India. Contrib. Miner. Petrol. 76, 420–429.

Sen, S.K., Bhattacharya, K., 1990. Granulites of Satnuru and Madras area: a study in different behaviors of fluids. In: In: Vielzeuf, D., Vidal, P. (Eds.), Granulites and Crustal Evolution, NATO ASI Series, Series C, vol. 311. pp. 367–384.

Srinivasan, R., Tareen, J.A.K., 1972. Andalusite from the Hospet area, Sandur Schist Belt, Mysore state. Indian Miner. 13, 42–45.

Srivastava, R.K., Hamilton, M.A., Jayananda, M., 2011. A 2.21 Ga large igneous province in the Dharwar Craton, India. In: International Symposium: Large Igneous Provinces of Asia; Mantle Plumes and Metallogeny. Abstract Volume, Irkutsk, Russia.

Stacey, J.S., Kramers, J.D., 1975. Approximation of terrestrial lead isotope evolution by two-stage model. Earth Planet. Sci. Lett. 26, 207–221.

Stähle, H.J., Raith, M., Hoernes, S., Delfs, A., 1987. Element mobility during incipient granulites formation at Kabbaldurga, Southern India. J. Petrol. 28, 803–834.

Swami Nath, J., Ramakrishnan, M., 1981. Early precambrian supracrustals of Southern Karnataka. Geol. Surv. IndiaMem. 112, 363p.

Zijderveld, J.D.A., 1967. Demagnetization of rocks: analysis of results. In: Collinson, D.C., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Palaeomagnetism. Elsevier, New York, pp. 254–286.

View publication stats